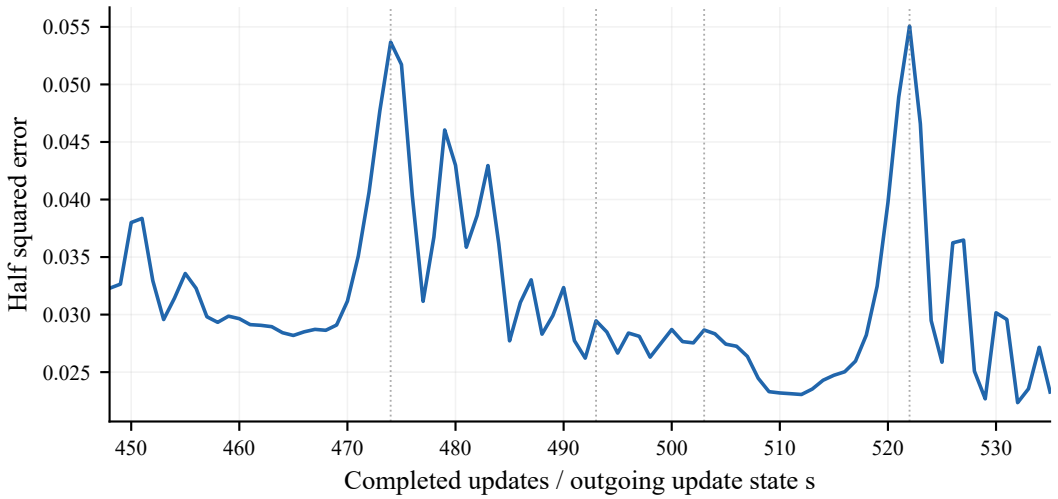
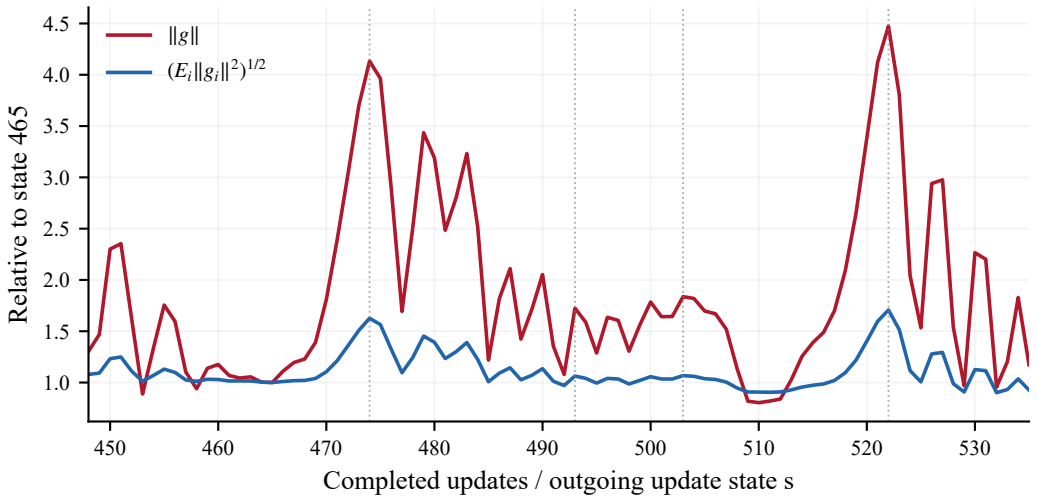
MLP Adam control: coherent gradient spikes without a normalization collapse

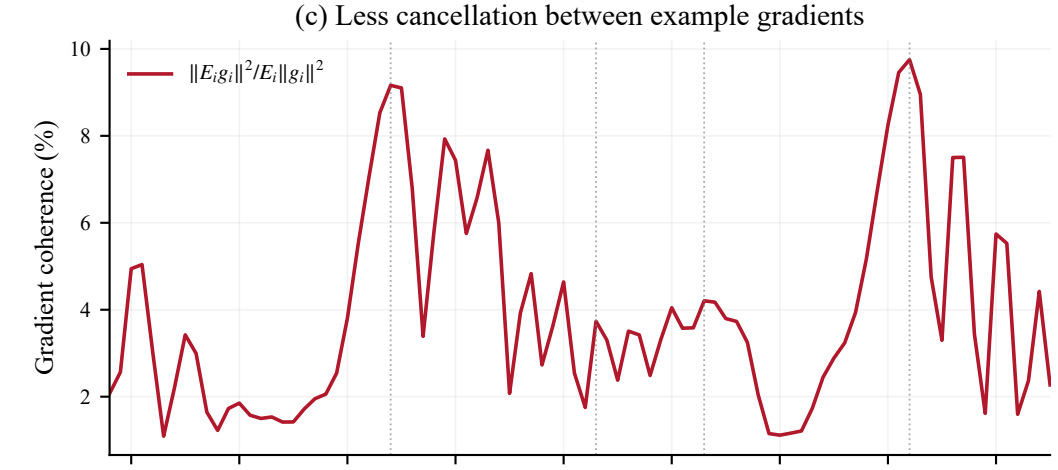
(a) Full training objective



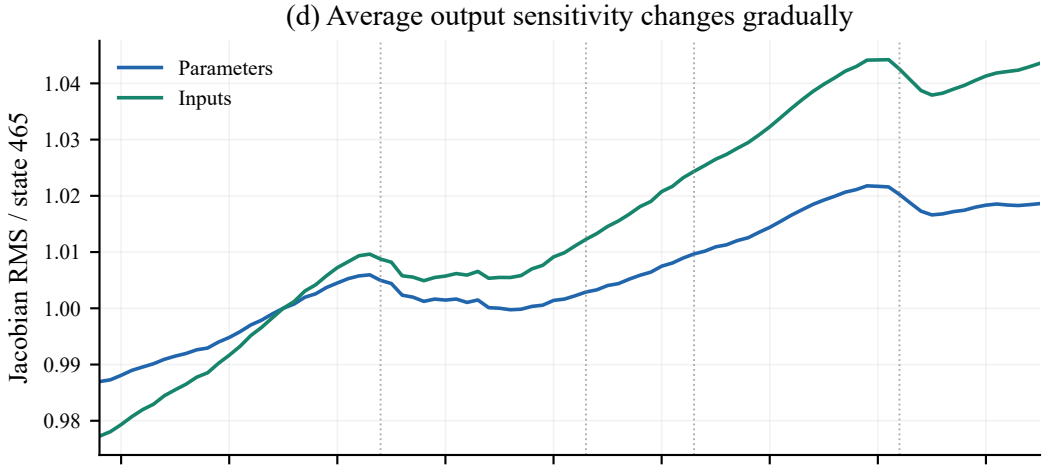
(b) Batch gradient grows faster than individual gradients



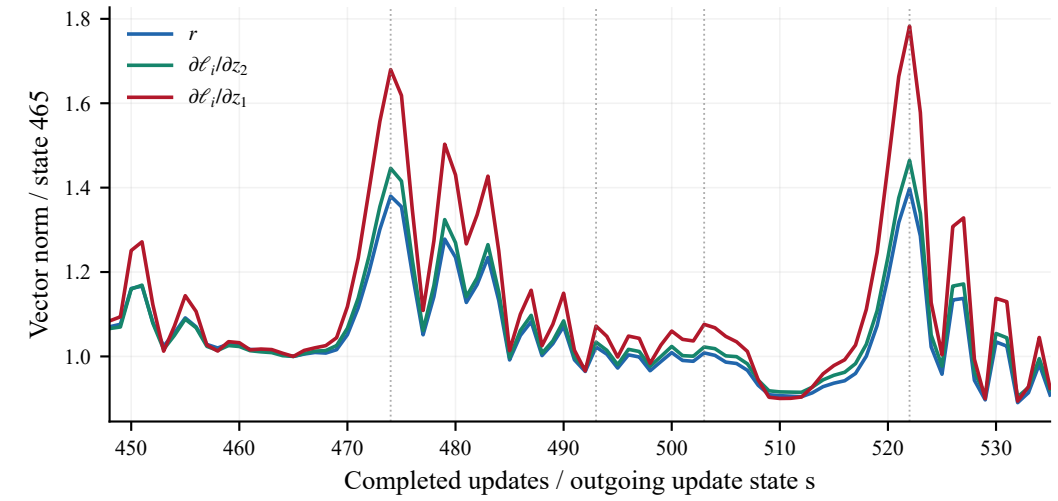
(c) Less cancellation between example gradients



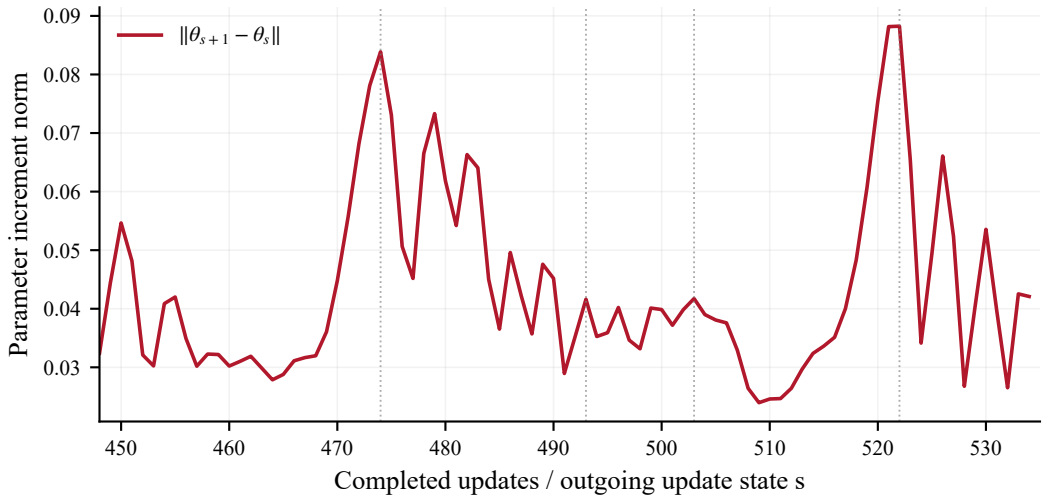
(d) Average output sensitivity changes gradually



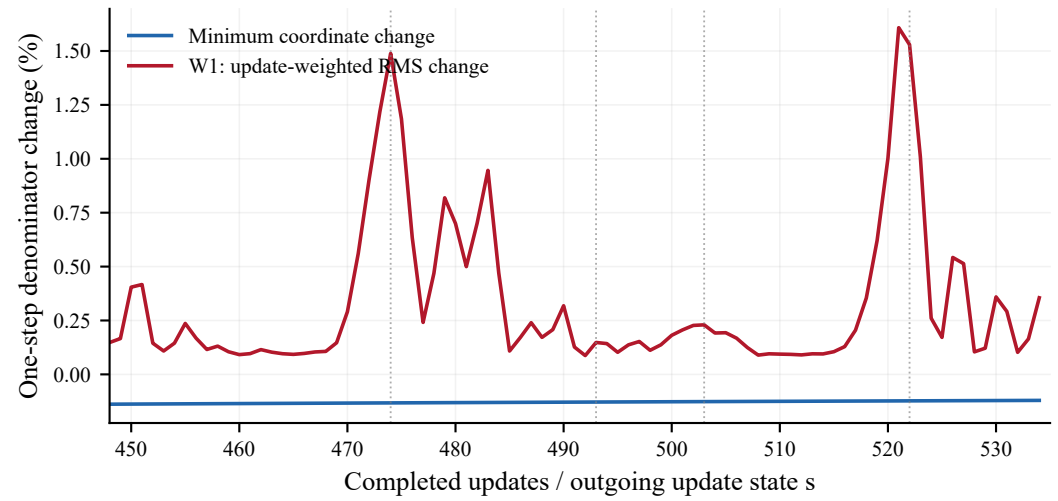
(e) Unit-loss backpropagation strengthens



(f) Actual Adam increments become larger



(g) Adam denominators do not collapse



(h) Successive prediction changes reverse direction

